

Guess, Check, Correct

Foundations of AI

Carlos Cotrini

UZH CAS Gamechanger AI

26 September 2026

About me



Dr. Carlos Cotrini

- BSc Mathematics, 2009
- MSc Information Security, ETH, 2014
- PhD Information Security, ETH, 2019
- PostDoc in AI and ML, ETH, 2019 to 2021
- Senior scientist, ETH, 2021 to now

Courses at ETH and UZH

- Advanced machine learning, 2019 to now
- Statistical learning theory, 2020 to 2022
- AI in medicine, 2020 to 2022
- Data analysis in physics, 2020 to 2025
- Stochastics and machine learning, 2024 to now
- Building AI/ML applications, 2024 to now
- AI and IT in industry, 2025 to now
- Software engineering fundamentals, 2025 to now
- AI project, 2026 to now
- Applied introduction to machine learning, from 2027

We study how we teach this

Transforming Confusion into Diffusion SIGCSE TS 2026

CER Best Paper (Global) award

- **Full stack ML:** you build the model from the ground up before you reach for a library.
- Trial with $N = 208$: that beat the library first approach by about **10 percent** on conceptual understanding ($p = 0.006$), and left participants more curious.

PATHOS: A Pedagogical Method for Sequencing Instruction in Multi-Foundational Machine Learning ICER 2026

- “...multi-foundational: they draw on multiple distinct fields and have complex prerequisite structures.”
- The order we teach things in is itself designed. Trial with $N = 209$: the designed order scored higher ($p = 0.011$, $d = 0.32$) at lower cognitive load.

Experimental-study team Dr. Sverrir Thorgeirsson · Stefan Cadar · Sofi Mousia · Umut Demirbas

Johannes Schulte-Vels · Nikolaj Murasov · Jonathan Maillefaud · Oscar Jauffret

Foundations of AI, UZH CAS Gamechanger AI, 26 September 2026

Transforming Confusion into Diffusion: Advancing Machine Learning Education via Bottom-Up Instruction

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Abstract

Reducing conceptual depth with practical skill development is a perennial challenge in advanced machine learning (ML) education, where powerful frameworks can obscure underlying mathematical and computational principles. To address this, we crafted a new pedagogical approach that we call full-stack machine learning (FSML), which emphasizes the construction of large language models and diffusion models from scratch. To evaluate the effectiveness of FSML, we conducted a classroom-based randomized controlled trial (RCT) in which FSML-based instruction was compared against a popular library-based instructional approach. We measured the study's conceptual understanding through a specialized assessment and administered a survey capturing knowledge gaps, resources, interests, and cognitive load. We found that students who received FSML instruction performed approximately 10% better than control participants on a quiz on transformers and stable diffusion ($p = 0.006$). They also showed increased curiosity and more positive affective responses, suggesting deeper engagement with ML fundamentals. Our findings indicate that our full-stack approach is effective in improving individual learning outcomes, previously including curriculum for ML and other advanced computing topics.

ACM Reference Format:
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1 Introduction

Computer science (CS) education has many dimensions that are relevant to their colleagues from other disciplines, one of which is how to balance high-level conceptual approaches to the subject with foundational skill development. While top-down approaches that involve collection and memorization of facts are the mainstream and dominant means of education [1, 10], they may leave students short of practical proficiency when not paired with consistent practice in hands-on skills—a position well known, for example, in foundational library education [15, 16] and mathematics [13, 14]. Consequently, many educators face a kind of top-down and bottom-up methods to syllabus on the strength of both, although the best way to integrate these is competing education remains a topic of ongoing discussion [19].

In machine learning (ML), this question is particularly

PATHOS: A Pedagogical Method for Sequencing Instruction in Multi-Foundational Machine Learning

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Abstract

Designing and ordering modern advanced topics in computer science like ML and diffusion models are multi-foundational: they draw on multiple distinct fields and have complex prerequisite structures. This makes instructional sequencing particularly challenging, yet no systematic method exists for designing presentation orders for such topics. We present PATHOS (Pedagogical Approach to Teaching Order Sequencing), an on-site sequential sequencing strategy that is designed to minimize extraneous material and enhance cognitive load for multi-foundational topics. We evaluated PATHOS in a pre-registered, randomized controlled trial ($N = 209$) in a university ML course, comparing a PATHOS-based approach for teaching diffusion models against a

library-based approach for teaching diffusion models against a

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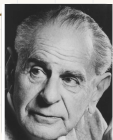
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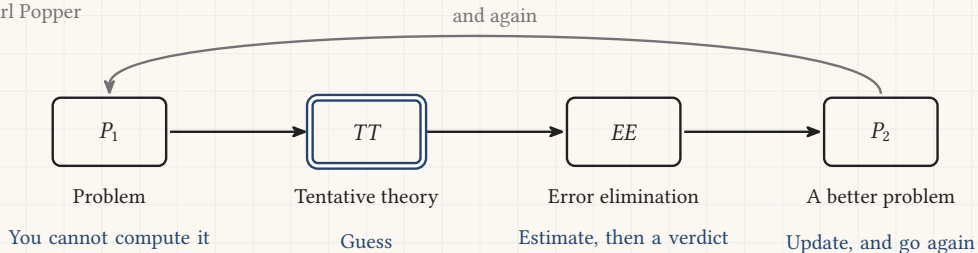
Popper, 1934: the tetradic schema



Karl Popper

We do not know: we can only guess.

Karl Popper, *The Logic of Scientific Discovery*, chapter 10, section 85 (1959).



This weekend is that diagram, and how it produces the AI applications you already use.

Schema after Karl Popper, *Objective Knowledge*, chapter 8 (1972).

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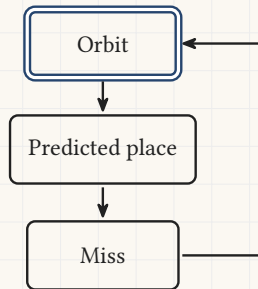
Astronomy: the recovery of Ceres, 1801



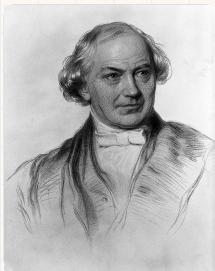
Carl Friedrich
Gauss, 1840.

C. A. Jensen

- **January 1801.** Giuseppe Piazzi finds a new object and tracks it for 41 days.
- **February 1801.** He loses it in the Sun's glare, with too few observations to fix an orbit.
- **Late 1801.** Carl Friedrich Gauss guesses an orbit, predicts, and corrects.
- **December 1801.** Franz Xaver von Zach finds it again.



Whewell, 1858: trying wrong guesses



William Whewell, stipple engraving.
Wellcome Collection, CC BY 4.0

To try **wrong guesses** is, with most persons, the only way to hit upon right ones.

The tendencies of our speculative nature ... perpetually show their vigour by **overshooting the mark**. They obtain something, by aiming at much more.

Most guesses are wrong, and a step that aims too far is how the right one gets found.

William Whewell, *Novum Organon Renovatum*, third edition (1858), book II, chapter V, page 79.

Computer science: Turing, 1950



Alan Turing, 29 March 1951.
Elliott and Fry

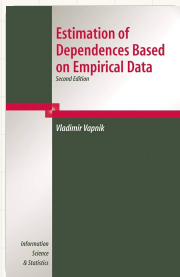
We cannot expect to find a good child-machine at the first attempt. One must experiment with teaching one such machine and see how well it learns. One can then **try another** and see if it is better or worse.

Now the learning process may be regarded as a **search for a form of behaviour** which will satisfy the teacher (or some other criterion).

The loop was written down as the plan for machine intelligence before there was a machine to run it on.

A. M. Turing, "Computing Machinery and Intelligence", *Mind* LIX(236), 1950, section 7.

The loop has a name, and the name is a book



Second edition, Springer, 2006

Vladimir Vapnik. The Russian original is 1979; Samuel Kotz's English translation is Springer, 1982.

The title is the whole claim. You are handed **empirical data**, and what you estimate from it is a **dependence**: the thing in the world that produced the numbers.

The book sets out **empirical risk minimisation**, which is the strategy underneath most of what the rest of this course will build.

Vladimir N. Vapnik, *Estimation of Dependences Based on Empirical Data*. Translated by Samuel Kotz. New York: Springer, 1982; second edition, Springer, 2006.

First-order approximation algorithms

Foundations of AI

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1

Section 1

Two problems, done by hand

Try it: Square One

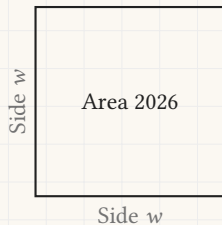
Switzerland, folded into a perfect square.

How long is its side?

<https://carloscotrini.github.io/FAI-PAGE/viz/square-root-show/>



Calculating square roots

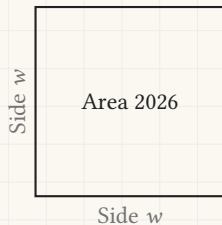


Find w with $w^2 = 2026$.

w	w^2	$l = 2026 - w^2$	Update
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How long is the side?

Calculating square roots

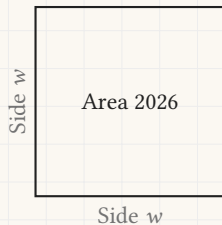


Find w with $w^2 = 2026$.

w	w^2	$l = 2026 - w^2$	Update
40	1600	+426	↑

How long is the side?

Calculating square roots

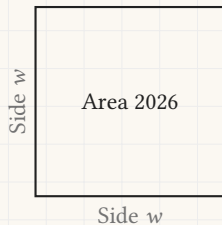


Find w with $w^2 = 2026$.

w	w^2	$l = 2026 - w^2$	Update
40	1600	+426	↑
50	2500	-474	↓

How long is the side?

Calculating square roots

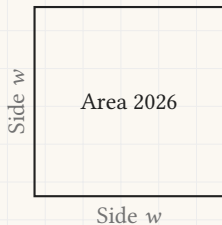


Find w with $w^2 = 2026$.

w	w^2	$l = 2026 - w^2$	Update
40	1600	+426	↑
50	2500	-474	↓
45	2025	+1	↑

How long is the side?

Calculating square roots

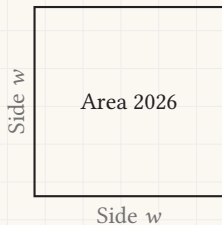


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⋮	⋮	⋮	

How long is the side?

Calculating square roots



How long is the side?

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w	w^2	$l = 2026 - w^2$	Update
40	1600	+426	↑
50	2500	-474	↓
45	2025	+1	↑
⋮	⋮	⋮	

Nobody here computed a square root. Each row is a guess that was scored, and the score said which way to move.

The update rule

$$w := w$$

$$\frac{w \quad w^2 \quad l}{\quad}$$

$$\frac{w \quad w^2 \quad l}{\quad}$$

The update rule

$$w := w + \underbrace{l}_{\text{The verdict}}$$

w	w^2	l
-----	-------	-----

w	w^2	l
-----	-------	-----

The update rule

$$w := w + \underbrace{\alpha}_{\text{How far to move}} \underbrace{l}_{\text{The verdict}}$$

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The update rule

$$w := w + \underbrace{\alpha}_{\text{How far to move}} \underbrace{l}_{\text{The verdict}}$$

w	w^2	l
40.000	1600.000	426.000
42.130	1774.937	251.063
43.385	1882.286	143.714
44.104	1945.153	80.847
44.508	1980.973	45.027
44.733	2001.064	24.936
44.858	2012.234	13.766
44.927	2018.414	7.586
44.965	2021.824	4.176
44.986	2023.702	2.298
44.997	2024.736	1.264
45.003	2025.305	0.695

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-----	-------	-----

- Starting at $w = 40$, with $\alpha = 0.005$.

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44.997	2024.736	1.264
45.003	2025.305	0.695

w	w^2	l
45.007	2025.618	0.382
45.009	2025.790	0.210
45.010	2025.884	0.116
45.010	2025.936	0.064
45.011	2025.965	0.035
45.011	2025.981	0.019
45.011	2025.989	0.011
45.011	2025.994	0.006
45.011	2025.997	0.003
45.011	2025.998	0.002
45.011	2025.999	0.001

- Starting at $w = 40$, with $\alpha = 0.005$.
- The whole run, all 23 rounds of it.
- From the fifth row of this block, w has stopped moving and the verdict is still shrinking.

The update rule

$$w := w + \underbrace{\alpha}_{\text{How far to move}} \underbrace{l}_{\text{The verdict}}$$

w	w^2	l
40.000	1600.000	426.000
42.130	1774.937	251.063
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45.011	2025.989	0.011
45.011	2025.994	0.006
45.011	2025.997	0.003
45.011	2025.998	0.002
45.011	2025.999	0.001

- Starting at $w = 40$, with $\alpha = 0.005$.
- The whole run, all 23 rounds of it.
- From the fifth row of this block, w has stopped moving and the verdict is still shrinking.
- $\sqrt{2026} = 45.011110$, and no square root was ever taken.

α is the step size. The letter ε is reserved: it is the stopping tolerance, and at 11:00 it becomes the width of a difference window.

Solving an equation

Find w with $w^3 - 3w^2 + 3w - 2 = 4$.

w	$f(w)$	$l = 4 - f(w)$	Update
-----	--------	----------------	--------

Solving an equation

Find w with $w^3 - 3w^2 + 3w - 2 = 4$.

w	$f(w)$	$l = 4 - f(w)$	Update
0	-2	+6	↑

Solving an equation

Find w with $w^3 - 3w^2 + 3w - 2 = 4$.

w	$f(w)$	$l = 4 - f(w)$	Update
0	-2	+6	↑
6	124	-120	↓

Solving an equation

Find w with $w^3 - 3w^2 + 3w - 2 = 4$.

w	$f(w)$	$l = 4 - f(w)$	Update
0	-2	+6	↑
6	124	-120	↓
2	0	+4	↑

Solving an equation

Find w with $w^3 - 3w^2 + 3w - 2 = 4$.

w	$f(w)$	$l = 4 - f(w)$	Update
0	-2	+6	↑
6	124	-120	↓
2	0	+4	↑
⋮	⋮	⋮	

The algorithm

```
def solve(f, target, w, alpha):  
    l = target - f(w)           # the verdict: how far off, and which way  
    while abs(l) > 0.001:       # not close enough yet  
        w = w + alpha * l      # the update: move that way  
        l = target - f(w)      # score the new guess  
    return w
```

The two problems, one call

The two problems of the last ten minutes are the same call, with a different f :

```
def square(w):  
    return w**2  
def cubic(w):  
    return w**3 - 3*w**2 + 3*w - 2  
solve(square, 2026, 40, 0.005)  
solve(cubic, 4, 0, 0.05)
```

2

Section 2

Iterative problem solving

Iterative problem solving

Iterative problem solving

Guess A value of w , however crude.

Guess

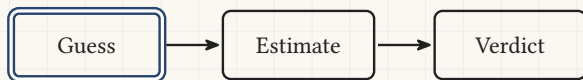
Iterative problem solving



Guess A value of w , however crude.

Estimate What the guess predicts.

Iterative problem solving

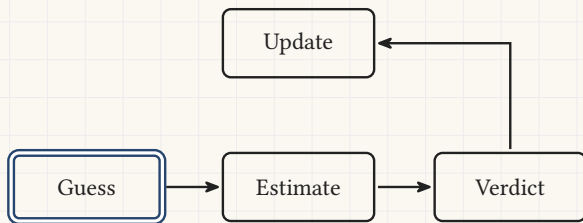


Guess A value of w , however crude.

Estimate What the guess predicts.

Verdict The signed miss l .

Iterative problem solving



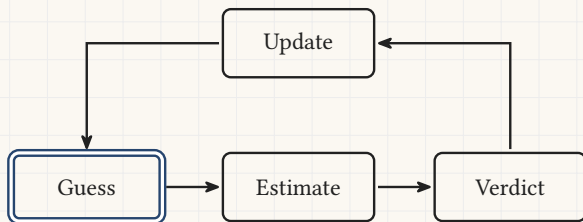
Guess A value of w , however crude.

Estimate What the guess predicts.

Verdict The signed miss l .

Update Move w the way the verdict says.

Iterative problem solving



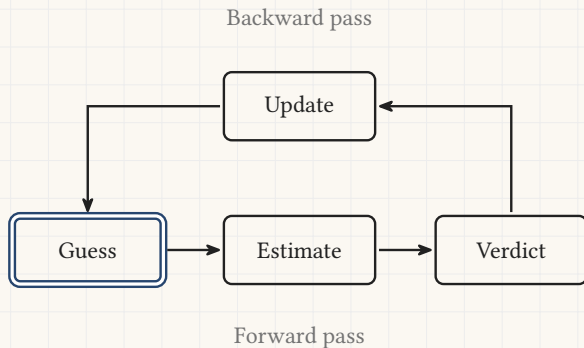
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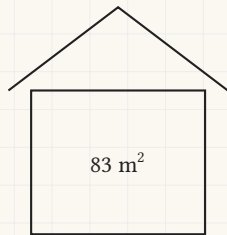
Update Move w the way the verdict says.

3

Section 3

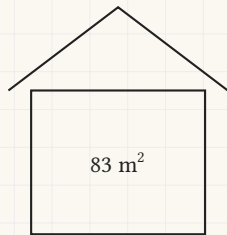
Estimating house prices

One house, one measurement



2 156 342 CHF

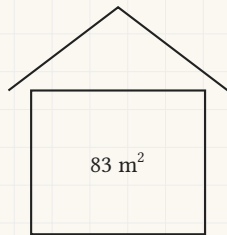
One house, one measurement



2 156 342 CHF

Data $x = 83$, the area. $y = 2\,156\,342$, the price.

One house, one measurement

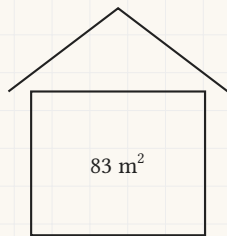


2 156 342 CHF

Data $x = 83$, the area. $y = 2\,156\,342$, the price.

Hypothesis Price is proportional to area, so the estimate is $\hat{y} = w x$.

One house, one measurement



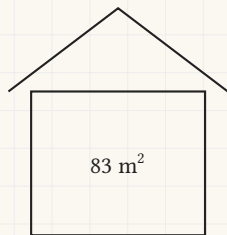
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Verdict The signed miss $l = y - \hat{y}$.

One house, one measurement



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Data $x = 83$, the area. $y = 2\,156\,342$, the price.

Hypothesis Price is proportional to area, so the estimate is $\hat{y} = w x$.

Verdict The signed miss $l = y - \hat{y}$.

Problem Find the w that drives l to zero.

One house: the loop

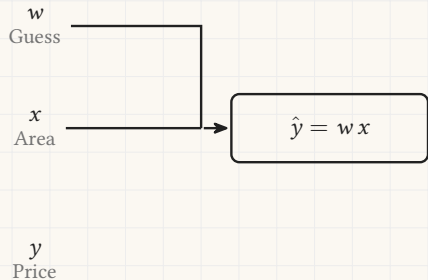
One house: the loop

w
Guess

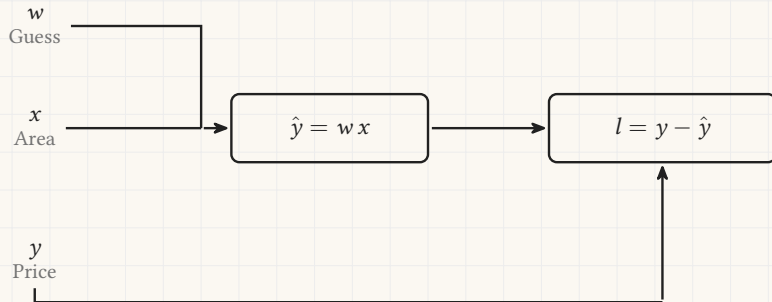
x
Area

y
Price

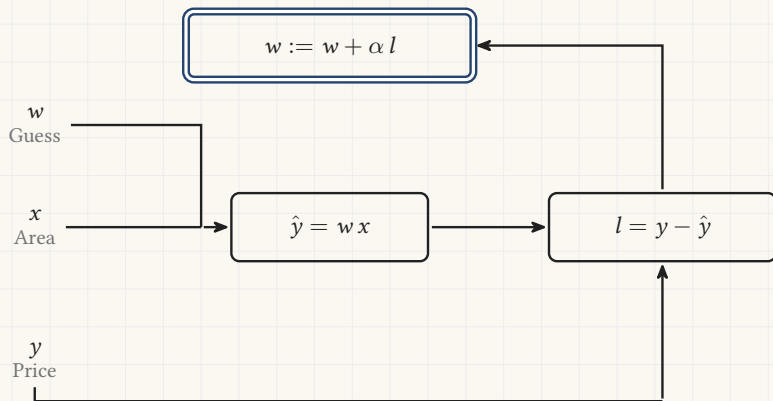
One house: the loop



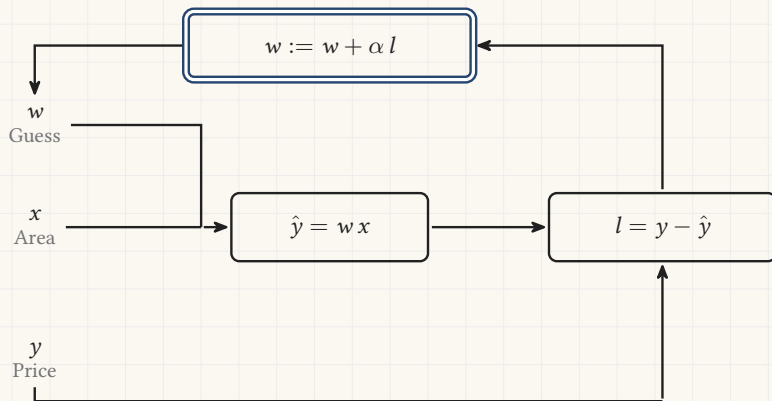
One house: the loop



One house: the loop



One house: the loop



One house: the algorithm

```
def fit(x, y, w, alpha):  
    l = y - w * x           # the verdict: y minus the estimate  
    while abs(l) > 0.5:      # more than half a franc out  
        w = w + alpha * l   # the update  
        l = y - w * x       # score the new guess  
    return w
```

One house: the algorithm

```
def fit(x, y, w, alpha):  
    l = y - w * x           # the verdict: y minus the estimate  
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        l = y - w * x       # score the new guess  
    return w
```

On the house above:

```
fit(83, 2156342, 20000, 0.01)
```

One house: the algorithm

```
def fit(x, y, w, alpha):  
    l = y - w * x           # the verdict: y minus the estimate  
    while abs(l) > 0.5:     # more than half a franc out  
        w = w + alpha * l  # the update  
        l = y - w * x      # score the new guess  
    return w
```

On the house above:

```
fit(83, 2156342, 20000, 0.01)
```

Starting from a guess of 20 000 francs per square metre, this returns 25 980.02 after 8 rounds.

Many houses, one measurement

House	Area	Price
-------	------	-------

Many houses, one measurement

House	Area	Price
1	83	2 156 342
2	62	1 461 635
3	104	2 650 166
4	128	3 184 899
5	145	3 651 113

House 1 is the house on the last slide. Houses 2 to 5 are synthetic, generated with a fixed seed.

Many houses, one measurement

Hypothesis The same one, row by row: $\hat{y}_i = w x_i$.

House	Area	Price
1	83	2 156 342
2	62	1 461 635
3	104	2 650 166
4	128	3 184 899
5	145	3 651 113

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Many houses, one measurement

House	Area	Price
1	83	2 156 342
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4	128	3 184 899
5	145	3 651 113

Hypothesis The same one, row by row: $\hat{y}_i = w x_i$.

Verdict One miss per row, $l_i = y_i - \hat{y}_i$.

House 1 is the house on the last slide. Houses 2 to 5 are synthetic, generated with a fixed seed.

Many houses, one measurement

House	Area	Price
1	83	2 156 342
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Problem No single w empties every row. Drive the *average* verdict $\bar{l}$ to zero instead.

$$\bar{l} = \frac{l_1 + l_2 + \dots + l_n}{n}$$

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Many houses: the loop

Many houses: the loop

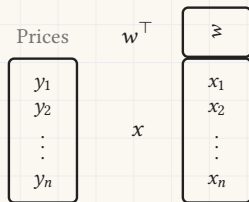
Prices

y_1
 y_2
 $\vdots$
 y_n

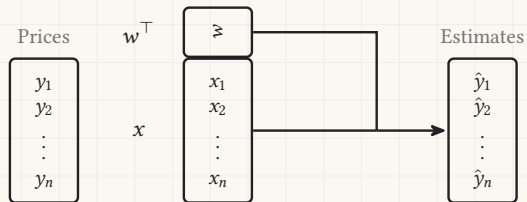
x

x_1
 x_2
 $\vdots$
 x_n

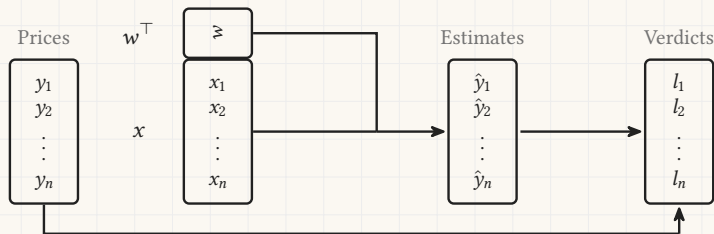
Many houses: the loop



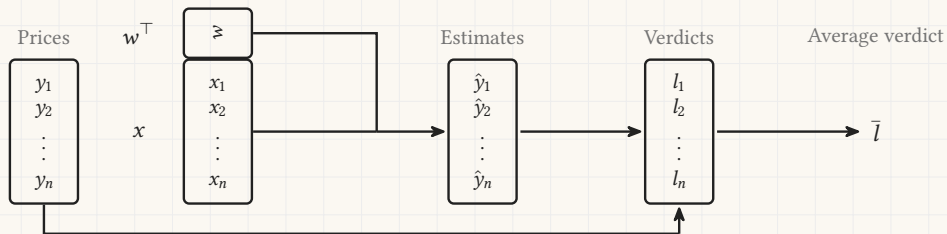
Many houses: the loop



Many houses: the loop

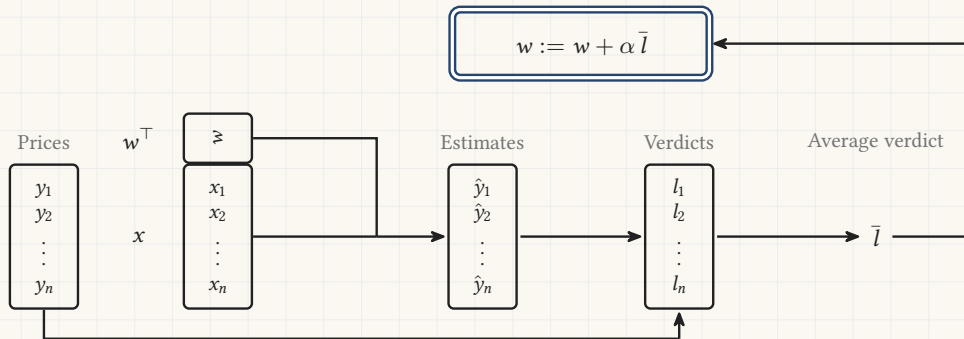


Many houses: the loop



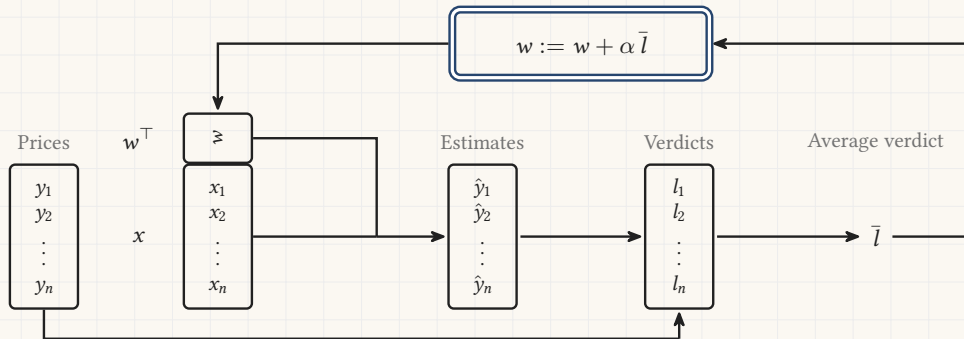
x holds the areas, and $\hat{y}_i = w x_i$, $l_i = y_i - \hat{y}_i$, $\bar{l} = (l_1 + \dots + l_n)/n$

Many houses: the loop



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Many houses: the algorithm

```
def verdict(x, y, w):           # the average miss over the rows
    total = 0
    for i in range(len(x)):
        total = total + (y[i] - w * x[i])
    return total / len(x)

def fit(x, y, w, alpha):
    l_bar = verdict(x, y, w)
    while abs(l_bar) > 0.5:
        w = w + alpha * l_bar
        l_bar = verdict(x, y, w)
    return w
```

Many measurements for each house

House	Area	Rooms	Age	Walk	Price

Many measurements for each house

House	Area	Rooms	Age	Walk	Price
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- One verdict, d unknowns. Pick one weight per round and move it.

Many measurements: the loop

Many measurements: the loop

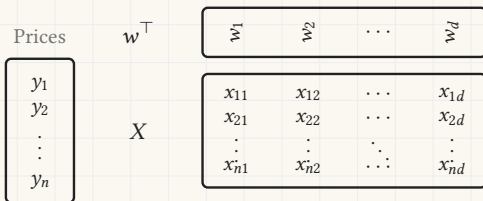
Prices

y_1
y_2
$\vdots$
y_n

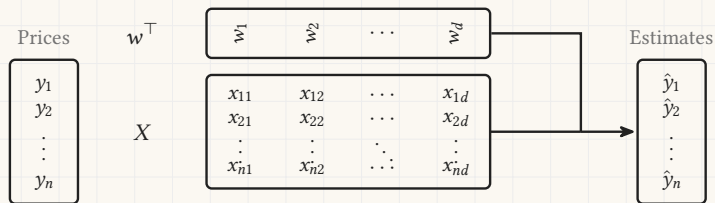
X

x_{11}	x_{12}	$\dots$	x_{1d}
x_{21}	x_{22}	$\dots$	x_{2d}
$\vdots$	$\vdots$	$\ddots$	$\vdots$
x_{n1}	x_{n2}	$\dots$	x_{nd}

Many measurements: the loop

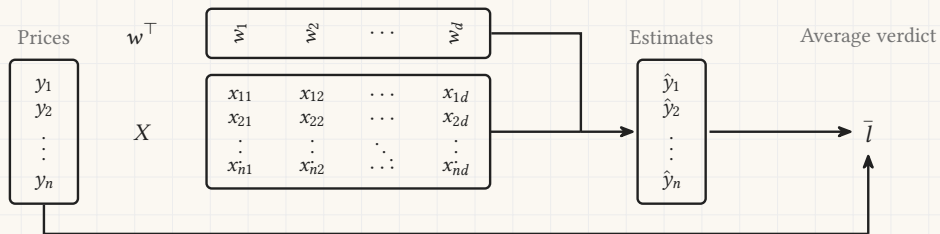


Many measurements: the loop



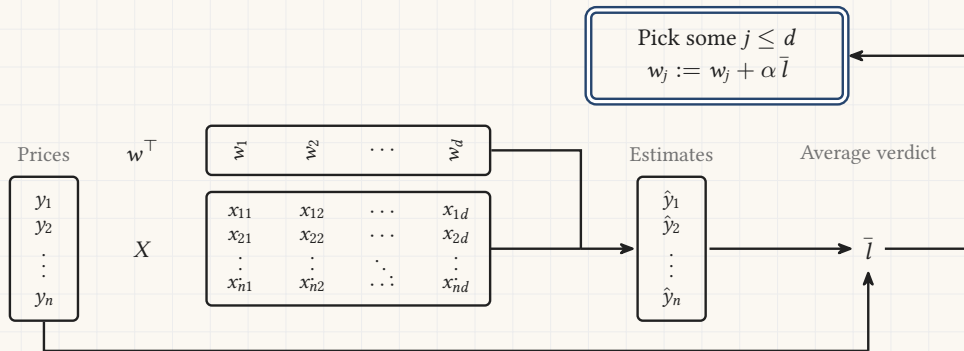
X has one row per house and one column per measurement, and $\hat{y}_i = w_1 x_{i1} + w_2 x_{i2} + \dots + w_d x_{id}$.

Many measurements: the loop



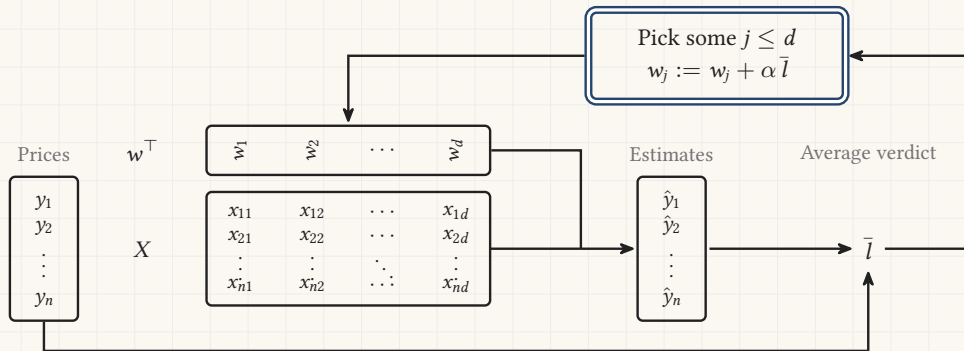
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